Does VAR Truncation Bias have implication on DSGE Structural Parameters when using IRF matching estimator? *

Lorenzo Del Col[ξ]

October, 2026

Abstract

Yes. Following De Graeve and Westermark (2025) and Baumeister (2025), I increase the lag length of the VAR in Altig et al. (2005) to re-estimate their DSGE. Moving from $p = 4$ to $p = 12$ lags, I find that both the VAR and DSGE impulse responses, and the implied structural parameters, change substantially. The Calvo price parameters that distinguish the observationally equivalent firm-specific and homogeneous capital versions of the model converge to one another, from 0.34 against 0.82 at $p = 4$ to 0.89 against 0.90 at $p = 12$. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters, equal to that of the homogeneous capital model, re-opening the micro–macro discrepancy in price stickiness that firm-specific capital was introduced to resolve.

Keywords: Truncation Bias, Long-Lag VARs, DSGE Estimation, IRF Minimum Distance , Nominal Rigidities
JEL Codes: C32, C52, E32

*...

[ξ]University of Padova. Contact: lorenzo.delcol@studenti.unipd.it

Contents

1 Introduction **1**

1.1 Motivation 1

1.2 Contribution and Results 2

1.3 Related Literature 3

1.4 Outline 3

2 Truncation Bias and the Case for Long-Lag VARs **4**

2.1 A simple illustration 4

2.2 Two channels: pure truncation and identification bias 4

2.3 Why longer lags help 5

2.4 Inference beyond the truncation horizon 6

3 Stochastic singularity and the absence of a VAR representation **6**

3.1 When does a DSGE admit a VAR representation? 6

3.2 How the literature has dealt with singularity 7

3.3 The present exercise: singular, and at the parametrisation frontier 8

4 The Theoretical Model **10**

4.1 Inflation inertia and the micro–macro conflict 10

4.2 Homogeneous and firm-specific capital 11

4.3 The role of χ 11

5 Econometric Methodology **12**

5.1 Impulse response matching 12

5.2 What the strategy rests on 13

6 Empirical Results **13**

6.1 Data and specification 13

6.2 VAR impulse responses: monetary policy shock 14

6.3 Technology shocks: short-run versus long-run identification 14

6.4 DSGE structural parameters 15

6.5 Implied duration and the collapse of the firm-specific mechanism 16

6.6 Residual diagnostics 18

7 Conclusion **18**

A Data **21**

B The ACEL Model: Full Derivation **21**

C Additional Residual Analysis **21**

D Monte Carlo Inference on the Calvo Parameters **26**

E Robustness **26**

F Replication Files and Code **26**

1 Introduction

1.1 Motivation

An important goal of monetary business cycle research is to explain the empirical evidence on the impact of economic shocks on macroeconomic variables. Because the systematic component of policy is endogenous, identifying the causal effect of monetary policy requires isolating the non-systematic component, the monetary policy shock, even though it is small and infrequent (Ravenna, 2007; Aruoba and Drechsel, 2024).

A large class of estimated DSGE models is taken to the data through this channel. Impulse Response Function Matching Estimation (IRFME) is a limited-information strategy that chooses structural parameters to minimise the distance between the impulse responses implied by the model and those estimated from an identified VAR (Christiano et al., 2005; Altig et al., 2011; Rossi et al., 2012). The strategy is attractive precisely because it is partial: the researcher takes a stand on the identification of a subset of shocks, and remains agnostic about the remaining dimensions of the data.

That attractiveness rests entirely on one maintained assumption: that the VAR delivers *consistent* estimates of the moments being matched. If the reduced-form VAR adequately captures the data generating process, impulse responses are optimal at all horizons; if the VAR is misspecified, the specification error propagates to the structural statistics and compounds at each horizon (Ramey, 2016; Canova and Fosso, 2025) Ravenna (2007) "The bias introduced by truncation can lead to bias in the identification of the structural shock." Altig et al. (2011) say as much themselves: "the quality of our estimation strategy depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks".[1]

Since a VAR(∞) cannot be estimated on a finite sample, researchers truncate. The resulting VAR(p) is misspecified, and increasingly so the smaller is p (De Graeve and Westermark, 2025; Barnichon et al., 2026; Ravenna, 2007). Ravenna (2007) "the problem facing economistis is whether the number of lags p to be included is reasonable given the lenght of economics time series over which VARs are estimated. When estimating a VAR consistent with DSGE business cycle models it is standard to assume that including few lags is sufficient to provide a reasonable approximation to the exact infinite order VAR representation! De Graeve and Westermark (2025) show that the conventional objection to longer lags, that the additional parametrisation makes confidence intervals prohibitively wide, is incomplete, because reducing misspecification reduces not only bias but also variance. Their conclusion is addressed explicitly to applied researchers: "it is up to the researcher to show that her structural results are robust to significant increases in the lag length, far beyond the lag lengths typically considered in current practice".

[1] Altig et al. (2011, footnote 10, p. 3).

[**TBD:** Chiudere la motivation con il ponte verso la domanda di ricerca: finora la letteratura ha studiato cosa la troncatura fa alle IRF, cioe' all'output finale dell'analisi. Ma quando il VAR e' l'*input* di una stima strutturale, il bias non si ferma li'. Aggiungere il riferimento a Baumeister (2025), che mostra come gli effetti reali di lungo periodo trovati in Jordà et al. (2024) siano verosimilmente un artefatto della troncatura: stesso spirito, ma su IRF, non su parametri strutturali.]

1.2 Contribution and Results

This paper asks what lag truncation does to *structural* estimates, not merely to impulse responses. I replicate Altig et al. (2011) — henceforth ACEL — and re-estimate their DSGE by IRF matching against VARs of increasing lag order.

The contribution is twofold. First, relative to De Graeve and Westermark (2025), I move the question from the reduced form to the structural parameters. Second, and more specifically, I do so in a setting their design deliberately excludes. De Graeve and Westermark (2025) state that "all our experiments are based on two-shock models and two-variable VARs", and that where a candidate model does not fit that framework — Smets and Wouters — "we consider the model with only monetary policy and preference shocks".[2] Their data generating processes are square and fundamental by construction, so that truncation bias is isolated from singularity and invertibility. ACEL's model is not: it confronts three structural shocks with a ten-variable VAR. [**TBD:** Verificare che nessun altro abbia gia' fatto questo esercizio su un modello stocasticamente singolare. Se confermato, e' il claim centrale.]

My main results are the following. *First*, the impulse responses to a monetary policy shock retain their sign and broad shape across lag orders, but the humps peak substantially later as p increases, and the point estimates remain inside the VAR(4) confidence bands. *Second*, the responses to the two technology shocks — identified with long-run rather than short-run restrictions — do not: several fall outside the baseline confidence set. This asymmetry is what Ravenna (2007) predicts, since long-run restrictions require the sum of the entire lag polynomial and truncation error accumulates in exactly that object. *Third*, the structural estimates move. The Phillips curve slope falls from 0.0402 to 0.0129 and becomes precisely estimated only at $p = 12$; the cost of varying capital utilisation collapses from 1.99 to 0.004; the investment adjustment cost roughly doubles. *Fourth*, and as a consequence, the implied Calvo price parameters of the firm-specific and homogeneous capital models converge, from 0.34 against 0.82 to 0.89 against 0.90. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters.

The last result overturns ACEL's central conclusion. Their claim is that only the firm-

[2] De Graeve and Westermark (2025, p. 10). They also note that "invertibility is never a problem; all the models we consider are fundamental".

specific capital model reconciles macro evidence on inflation inertia with micro evidence on the frequency of price adjustment. Under a richer lag structure that reconciliation disappears. The channel is transparent and internal to their own argument: ACEL note that "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p".[3] Long-lag VARs push the estimate of σ_a precisely into that region.

1.3 Related Literature

Long-lag VARs. [**TBD:** Paragrafo su De Graeve and Westermark (2025), Baumeister (2025), Antolín-Díaz and Surico (2025), Antolín-Díaz et al. (2026). Cosa fanno: stabiliscono il caso econometrico per lag lunghi e ne mostrano le conseguenze sulle IRF. Contributo mio: ne traccio le conseguenze fino ai parametri strutturali, e in un modello stocasticamente singolare che i loro esperimenti escludono per costruzione.]

Truncation bias and the DSGE–VAR mapping. [**TBD:** Paragrafo su Ravenna (2007), Chari et al. (2008), Fernández-Villaverde et al. (2005), Forni et al. (2016), Giacomini (2013). Cosa fanno: caratterizzano quando un DSGE ammette una rappresentazione VAR finita e cosa costa approssimarla. Contributo mio: quantifico il costo di sbagliare p in un modello stimato di riferimento, non in un esempio calibrato.]

Impulse response matching and DSGE estimation. [**TBD:** Paragrafo su Christiano et al. (2005), Altig et al. (2011), Rossi et al. (2012), Theodoridis (2011). Cosa fanno: sviluppano e valutano la strategia limited-information. Contributo mio: mostro che la sua pretesa robustezza poggia su un'assunzione — la consistenza del VAR — che la troncatura viola sistematicamente. Ravenna (2007) "In the case of zero identification bias, Fig. 4 shows that the impulse response function from a correctly identified VAR(p) with p! 6 and 12 is accurate up to the pth lag (the error depicted in the plot converges to zero as the approximating VAR lag order p becomes large)." h¡p before Plagborg-Møller and Wolf (2021)]

[**TBD:** Valutare se serve un quarto blocco su LP contro VAR (Plagborg-Møller and Wolf (2021), Li et al. (2024), Ludwig (2026), Olea et al. (2025)) oppure se basta il rimando in Sezione 2.4, dove serve davvero. Con una pagina di budget, probabilmente basta il rimando.]

1.4 Outline

Section 2 illustrates the truncation bias problem, separates its two channels following Ravenna (2007), and reviews the case for longer lags. Section 4 presents the ACEL model and the role of firm-specific capital in reconciling micro and macro evidence on price

[3] Altig et al. (2011, p. 227).

setting. Section 5 describes the econometric strategy: why a stochastically singular DSGE is estimated by limited-information methods, and what that strategy assumes. Section 6 presents the empirical results. Section 7 concludes.

2 Truncation Bias and the Case for Long-Lag VARs

VARs are subject to at least three sources of bias: small-sample bias, identification bias (Forni et al., 2016), and truncation bias (De Graeve and Westermark, 2025; Ravenna, 2007; Chari et al., 2008; Barnichon et al., 2026). This paper is concerned with the latter, and with the way it contaminates the second.

2.1 A simple illustration

Following De Graeve and Westermark (2025), consider a data generating process in which y_t depends on two exogenous variables,

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t. \tag{1}$$

Estimating the relationship while omitting x_{2t},

$$y_t = x_{1t}b_1 + e_t, \tag{2}$$

delivers a biased point estimate unless the regressors are orthogonal, $\mathbb{E}(b_1) \neq \beta_1$, and an upward-biased residual variance, $s^2 > \sigma^2$. Both distortions matter here, and the second is the one usually forgotten.

The same logic carries over to a VAR. Let

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \tag{3}$$

with $B(L) = B_1 L + \cdots + B_p L^p$ and $p \ll p^*$, where p^* denotes the true lag order. Truncation biases both the autoregressive coefficients $B(L)$ and the reduced-form covariance matrix Σ_u. Since impulse responses are a function of both,

$$\text{IRF} = f\big(B(L), \Sigma_u\big), \tag{4}$$

they inherit the bias.

2.2 Two channels: pure truncation and identification bias

Ravenna (2007) shows that the distortion operates through two separate channels. The first is *pure truncation bias*: a VAR(p) restricts all coefficients beyond lag p to zero and therefore

does not describe the true dynamics of the model. The second is *identification bias*: when the autoregressive coefficients enter the matrix mapping reduced-form innovations into structural shocks, truncation corrupts that mapping. Crucially, the identification bias arises *even when the identifying assumptions are correct*, and both results are established in population, so that neither is an artefact of small samples.

Two implications matter for what follows. First, Ravenna (2007) finds the identification channel to be the dominant one. Second, the two channels do not affect all identification schemes equally. Schemes based on long-run restrictions require the sum $\sum_{i=0}^{\infty} \Psi_i$ of the entire lag polynomial, and that sum can be estimated very poorly even when the individual coefficients are estimated tightly.[4] Ravenna (2007) documents parametrisations in which the VAR(p) approximates the infinite-order representation well coefficient by coefficient and nonetheless tracks the long-run identified response poorly. This asymmetry between short-run and long-run identification generates a testable prediction, which I take to the data in Section 6.3.

2.3 Why longer lags help

De Graeve and Westermark (2025) explore the obvious remedy, increasing p substantially, and find evidence against the conventional wisdom that uncertainty becomes prohibitive. The standard intuition is only part of the story: raising the lag order reduces misspecification, and reducing misspecification lowers not only bias but also the variance of the residuals. This variance reduction works against the imprecision induced by the additional parameters, and in their simulations on realistically sized samples it frequently dominates. The result is reduced truncation bias, more precise inference, lower mean squared error and better coverage.

A corollary concerns model selection. Information criteria are built to optimise one-step-ahead forecasting, an out-of-sample objective that rewards parsimony; structural inference is a different problem. De Graeve and Westermark (2025) put it directly: "information criteria focused purely on forecasting are poor guides for accurate structural inference".[5] This matters here because ACEL report that the Akaike, Hannan–Quinn and Schwartz criteria support $q = 2, 2, 1$ respectively — standard criteria point towards *fewer* lags than even their own baseline of four.

[**TBD:** Affrontare qui, di petto, il punto debole dell'esercizio, prima che lo faccia il lettore: RMSE in-sample e generalized variance migliorano meccanicamente all'aumentare di p, quindi non costituiscono prova di specificazione migliore. La posizione da tenere e' che non sostengo che $p = 12$ sia il valore "vero": sostengo che le conclusioni dipendono

[4] Ravenna (2007); see also the discussion in Christiano et al. (2006). [**TBD:** Aggiungere a biblio.bib: Sims (1972), Faust and Leeper (1997), Erceg, Guerrieri and Gust (2005). Gli ultimi due sono gia' nella cartella "Letteratura x Tesi".]

[5] De Graeve and Westermark (2025, p. 6).

da p e che questa sensibilita' vada riportata come prassi. Questo elimina il problema di model selection invece di provare a risolverlo.]

2.4 Inference beyond the truncation horizon

A third result of De Graeve and Westermark (2025) bears directly on how impulse responses should be read. They argue that a VAR(p) should not be used for structural inference at horizons $h > p$, because a VAR(p) cannot propagate much beyond horizon p: uncertainty is then mechanically underestimated, and the researcher risks becoming confident about the wrong point estimate. They also stress that truncation affects inference at *all* horizons, including $h \leq p$.

This has an uncomfortable implication for the exercise being replicated. ACEL's headline stylised facts are hump-shaped responses peaking roughly one to two years after the shock, that is at $h \approx 4$ to 8. With $p = 4$, essentially all of those horizons lie beyond the truncation horizon. The empirical target against which the DSGE is matched therefore lies in the region where the VAR is extrapolating through an estimated lag polynomial rather than estimating. I make this visible in Section 6 by marking $h = p$ in the impulse response figures.

[**TBD:** Collegare brevemente a Plagborg-Møller and Wolf (2021) e Ludwig (2026): LP e VAR stimano lo stesso estimando, e le LP sono combinazioni di VAR di ordine crescente, quindi confrontare LP e VAR a lag fissato mette a confronto un modello grande e uno piccolo. Tre o quattro righe, non di piu'.]

3 Stochastic singularity and the absence of a VAR representation

3.1 When does a DSGE admit a VAR representation?

The log-linearised solution of a DSGE model can be written in state-space (or ABCD) form, and Fernández-Villaverde et al. (2005) provide a simple condition — the *Poor Man's Invertibility Condition* (PMC) — under which that state-space system can be inverted into a VAR(∞) in the observables. The condition is that the eigenvalues of $A - BD^{-1}C$ lie inside the unit circle. The notation already reveals the maintained assumption: D^{-1} must exist, so D must be square. In economic terms, the model must have exactly as many structural shocks as observable variables.

This is not a technicality that can be waved away. When the number of shocks falls short of the number of observables the system is *stochastically singular*: a K-dimensional process driven by $r < K$ shocks has a spectral density of rank r, its Wold representation is $y_t = \Psi(L)\varepsilon_t$ with Ψ of dimension $K \times r$, and there is no non-singular K-dimensional

white noise. The reduced-form covariance matrix Σ_u is singular, Σ_u^{-1} does not exist, and a VARMA or VAR(∞) representation in the standard sense is not defined. Ravenna (2007) states the practical implication directly: when the number of shocks is smaller than the number of observables the system is singular, "preventing likelihood estimation of the VAR", and recovering a non-singular VAR representation requires dropping observables until the two counts coincide.

3.2 How the literature has dealt with singularity

De Graeve and Westermark (2025) generate artificial data from three well-known DSGE models — the RBC models of Chari et al. (2008) and Christiano et al. (2006), and Smets and Wouters (2007) — and estimate VARs on that data across short and long lag choices. Their design is explicit about the restriction just described: "invertibility is never a problem; all the models we consider are fundamental", and "all our experiments are based on two-shock models and two-variable VARs".[6] The two RBC models satisfy this by construction. The Smets–Wouters model does not: it carries seven structural shocks, so confronting it with a two-variable VAR would leave the system singular. Their solution is to reduce the model, retaining "only monetary policy and preference shocks", until the number of shocks matches the number of variables in the VAR.[7] In either direction, the operation is the same: squareness is imposed so that truncation bias can be studied in isolation from singularity and invertibility.

There is, however, a strand of the literature arguing that this requirement is far stronger than the analysis actually needs. Forni et al. (2016) observe that "fundamentalness is extremely and unnecessarily restrictive", because "in several cases fundamentalness is not needed. The researcher is often interested in assessing the transmission mechanisms of a single shock." What matters, in their view, "is not whether we have fundamentalness or not, but whether we have enough information to get a reasonably good approximation for the specific shocks of interest". They also note the cost of insisting on it: "if the number of variables in the VAR is smaller than the number of shocks in the DSGE, fundamentalness cannot hold (and the PMC is not even defined). Since modern DSGE models have several shocks, small VARs are automatically ruled out. This is unpleasant, since small VARs are useful tools, especially for short samples."

This is precisely the position that IRF matching occupies. The strategy does not require the DSGE to be the data generating process for all observables: it requires the VAR to be informative about the transmission of the shocks being matched. Section 5

[6] De Graeve and Westermark (2025, p. 10).

[7] Note the direction of the adjustment: the binding constraint is the size of the VAR laboratory, which is fixed at two variables throughout, so squareness is restored by discarding *shocks*, not observables. Discarding observables is the alternative route, and is the one suggested by Fernández-Villaverde et al. (2005) and Ravenna (2007).

develops this point.

3.3 The present exercise: singular, and at the parametrisation frontier

The VAR in Altig et al. (2011) confronts three structural shocks — monetary policy, neutral technology and capital-embodied technology — with ten observable variables. The system is therefore stochastically singular, the PMC is not defined for it, and there is no exact VAR(∞) representation of the DSGE that the VAR is being used to estimate. This is the first respect in which the present exercise departs from De Graeve and Westermark (2025): rather than restoring squareness by design, I study the consequences of lag truncation in a setting where singularity is a feature of the model being estimated and cannot be assumed away.

The second departure is one of scale, and it works in the opposite direction to what one might expect. Because the three data generating processes in De Graeve and Westermark (2025) are two-variable systems, their VARs remain cheaply parametrised even at long lag lengths, and increasing p costs relatively little in degrees of freedom. The ACEL VAR does not enjoy that luxury. With $K = 10$ variables, a constant and p lags, each equation carries $10p + 1$ coefficients, and the sample runs from 1959:Q2 to 2001:Q4, that is 171 quarterly observations before lags are consumed. Table 1 reports the resulting budget.

Table 1: Degrees of freedom across lag orders

	VAR $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Usable observations T	167	163	159
Coefficients per equation k	41	81	121
Degrees of freedom $T - k$	126	82	38
Observations per parameter T/k	4.07	2.01	1.31
Coefficients, full system	410	810	1,210

Note: Each equation contains $10p + 1$ coefficients ($K = 10$ variables, p lags, one constant). The raw sample contains 171 quarterly observations (1959:Q2–2001:Q4), of which p are consumed as initial conditions. Ordinary least squares requires $10p + 1 < 171 - p$, that is $p \leq 15$; at $p = 15$ only five degrees of freedom per equation remain.

At $p = 12$ the system carries 1,210 coefficients, each equation is estimated on 159 observations with 121 regressors, and only 38 degrees of freedom survive. The exercise is close to the algebraic ceiling: ordinary least squares requires $10p + 1 < 171 - p$, so no lag order above $p = 15$ is feasible at all, and $p = 15$ would leave five degrees of freedom per equation. Tripling the lag length relative to the ACEL baseline is therefore not a conservative perturbation — it is very nearly the largest perturbation the data permit.

This makes the exercise a demanding test of the central claim in De Graeve and Westermark (2025). Their argument is that longer lags reduce misspecification, that

reduced misspecification lowers the variance of the residuals, and that this variance reduction can offset the imprecision caused by additional parameters. In a two-variable system the offset has ample room to operate. Here it does not: the parametrisation grows by a factor of three in a system already large relative to the sample. If confidence intervals were going to explode anywhere, they would explode here.

They do not. Figure 1 plots the impulse responses to a monetary policy shock at each lag order together with their own 95 percent bootstrap confidence bands.

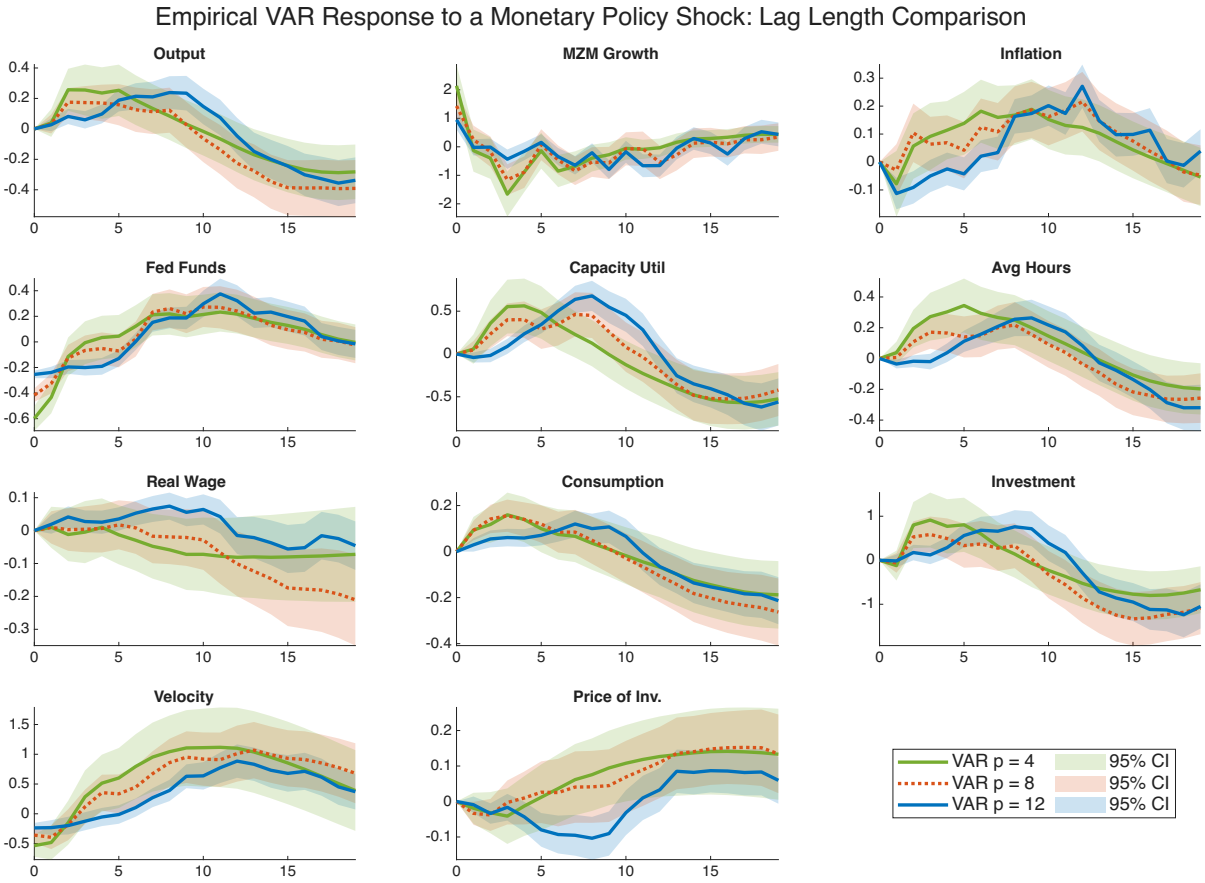


Figure 1: VAR impulse responses to a monetary policy shock across lag orders, each with its own 95 percent bootstrap confidence band.

The bands at $p = 12$ are not wider than those at $p = 4$; for several variables — inflation, the federal funds rate, the real wage, velocity and the relative price of investment — they are visibly narrower, despite the threefold increase in parametrisation documented in Table 1. The same pattern appears in the structural estimates: the Phillips curve slope is estimated roughly six times more precisely at $p = 12$ than at $p = 4$ (Section 6.4). The variance-reduction mechanism of De Graeve and Westermark (2025) therefore operates even at the parametrisation frontier, which is a considerably more demanding test than the two-variable systems in which it was originally demonstrated.

[**TBD:** Quantificare invece di limitarsi a "visibly narrower": calcolare la larghezza media della banda per variabile e lag order, e riportare il rapporto medio $p = 12$ contro

$p = 4$. Serve una riga di numeri nel testo, non solo il rimando alla figura.]

Two features of how these bands are constructed matter for interpreting them. First, they are bootstrap bands in which *the entire VAR is re-estimated on each artificial sample*, so the uncertainty induced by estimating 1,210 coefficients is fully propagated into the reported intervals rather than being conditioned away. Had the estimator been unstable at $p = 12$, the re-estimated responses would have varied wildly across draws and the bands would have widened accordingly. Second, draws yielding an explosive companion matrix are discarded and redrawn, so the bands condition on stationarity. [**TBD:** Riportare la frequenza di draw esplosivi (`count_var_expl` in `mkimplrnewci.m`) ai tre lag order: e' una diagnostica di stabilita' gratuita e un lettore attento la chiedera'.]

An obvious objection is that this is simply overfitting: a VAR with 121 regressors per equation fits the sample closely, the estimated residuals shrink, and confidence bands computed from those residuals shrink with them. The objection has to be taken seriously, because in-sample it cannot be dismissed — reduced misspecification and overfitting have the same in-sample signature, namely a smaller residual variance. What can be said here is that the bootstrap re-estimation described above rules out the mechanical part of the concern, since parameter uncertainty is not being suppressed. What it cannot rule out is that the residual variance itself is understated, because the bootstrap resamples from the fitted residuals of the very model under scrutiny. Distinguishing the two requires evidence that is not in-sample: this is precisely what De Graeve and Westermark (2025) supply, through coverage rates computed in a controlled environment where the true responses are known, and what motivates the bias–variance discussion in Section 2.3.

4 The Theoretical Model

4.1 Inflation inertia and the micro–macro conflict

Altig et al. (2011) start from a tension in the New Keynesian literature. To account for inertia in inflation, standard models require firms to re-optimise prices infrequently — every six quarters, and in versions without indexation to past inflation even every two years. This sits uneasily with the microeconometric evidence, which finds price changes considerably more frequent. [**TBD:** Citare Bils and Klenow (2004), Golosov and Lucas (2007), Klenow and Kryvtsov (2008), che sono i riferimenti di ACEL. Valutare se aggiungere evidenza micro piu' recente. Da aggiungere a biblio.bib.]

Altig et al. (2005) propose a resolution. In their model inflation is inertial even though firms re-optimise on average every 1.5 quarters. Inertia arises not because firms change prices rarely, but because when they do they change them by a small amount: each firm's short-run marginal cost is increasing in its own output, which dampens the optimal price adjustment. The assumption delivering this is that capital is predetermined within the

period and firm-specific, so that factors of production cannot be instantly reallocated across firms.

4.2 Homogeneous and firm-specific capital

ACEL conduct the analysis using two versions of the Christiano et al. (2005) model: one in which capital is homogeneous, as in standard formulations, and one in which it is firm-specific. Once log-linearised, the only difference between the two sets of equilibrium conditions lies in the equation relating inflation to marginal cost,

$$\Delta\hat{\pi}_t = \mathbb{E}\big[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t \mid \Omega_t\big], \tag{5}$$

where $\hat{s}_t$ is economy-wide average real marginal cost in units of the final good, and the slope γ maps into the structural parameters as

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\,\chi. \tag{6}$$

The information set Ω_t contains the current realisation of the technology shocks but not that of the monetary policy innovation.

The two models are parametrised in terms of γ rather than ξ_p. Given the remaining parameters they are then observationally equivalent with respect to aggregate prices and quantities, so that no stand need be taken on which version is being estimated. What distinguishes them is the value of ξ_p that a given γ implies.

4.3 The role of χ

Altig et al. (2011) establish that $\chi = 1$ in the homogeneous capital model, while in the firm-specific capital model χ is a particular non-linear function of the structural parameters.[8] Two forces push χ below one, and both work by making a firm move its price *less* when it is able to reset it:

- the costlier it is to vary capital utilisation, the steeper the firm's own marginal cost curve, and hence the smaller the optimal price adjustment;
- the more elastic the firm's demand curve, the larger the fall in output — and therefore in marginal cost — following a price increase.

In short, χ governs *how much* a firm moves its price and ξ_p governs *how often*, with implied duration $1/(1-\xi_p)$. Since γ is pinned down by aggregate data, a smaller χ

[8] Proposition 1 in Altig et al. (2011). [**TBD:** La derivazione e' in ACEL (2004) o in Christiano (2004): verificare quale sia il riferimento corretto e aggiungerlo a biblio.bib. Il PDF di Christiano (2004) e' in "Letteratura x Tesi".]

requires a smaller ξ_p: this is the sense in which firm-specific capital rationalises frequent price re-optimisation.

Two features of equation (6) will matter for the results. First, the mapping from γ to ξ_p is a quadratic that becomes nearly singular as $\gamma \to 0$, so that small changes in the estimated slope translate into large changes in the implied Calvo parameter, and normality does not survive the transformation. Second, and more substantively, ACEL themselves identify the fragility of their own mechanism: "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p".[9] Because the price markup λ_f is calibrated rather than estimated, the cost of varying capital utilisation σ_a is the only estimated parameter entering χ, and therefore the natural suspect whenever the gap between the two models moves.

[**TBD:** Decidere se calcolare $\hat{\chi}$ esplicitamente ai tre lag order e riportarlo in tabella. Se $\hat{\chi} \to 1$ passando da $p = 4$ a $p = 12$, quella diventa la tabella centrale del paper e la Sezione 6 si riorganizza attorno a essa.]

5 Econometric Methodology

5.1 Impulse response matching

Under full-information estimation the DSGE is written in state-space form and the likelihood of the raw data is evaluated through the Kalman filter. The maintained assumption is that the model *is* the data generating process: misspecification in any single block contaminates all parameter estimates, because the criterion weights the entire covariance structure. It also requires the system to be non-singular, which is why models estimated this way are built with as many shocks as observables.

IRFME makes a weaker claim. The moments matched are the responses of the observables to the identified shocks; the remaining dimensions of the innovation space are left unmodelled. The statement is not "my model generated the data" but "whatever generated the data, the response to this shock looks like this, and my model should reproduce it". The estimator chooses θ to minimise

$$\big[\hat{\Psi} - \Psi(\theta)\big]' W \big[\hat{\Psi} - \Psi(\theta)\big], \tag{7}$$

where $\hat{\Psi}$ collects the VAR-based impulse responses over the matched horizons, $\Psi(\theta)$ their model counterparts, and W is a weighting matrix. [**TBD:** Specificare quali orizzonti e quali variabili ACEL matchano e come e' costruito W (dalla matrice di covarianza delle IRF stimate). Recuperare dal replication package.]

Identification in the VAR is chosen to be consistent with the model. The monetary

[9] Altig et al. (2011, p. 227).

policy shock is identified through short-run recursive restrictions: only a subset of contemporaneous variables enters the information set of the policy rule. The two technology shocks are identified through long-run restrictions: technology innovations are the only shocks affecting the level of labour productivity in the long run, and capital-embodied technology shocks are the only ones affecting the relative price of investment in the long run. [**TBD:** Aggiungere Fisher (2006) a biblio.bib, che e' il riferimento seguito da ACEL.] There is no circularity in using the model to justify the restrictions and the restrictions to estimate the model, because the restrictions are qualitative and hold across a wide class of models and parameter values, while the parameters being estimated are quantitative.

5.2 What the strategy rests on

The robustness that IRFME buys is robustness to misspecification in the dimensions the researcher declines to model. It is not robustness to misspecification of the VAR itself. The strategy has one load-bearing assumption — that $\hat{\Psi}$ consistently estimates the population impulse responses — and lag truncation violates exactly that.

The consequence is worth being precise about. The estimator remains consistent, but for the pseudo-true parameter value that reconciles the model with the *biased* target. This is not an inference problem that wider standard errors would address: it is a displaced objective. Moreover, the identification channel of Ravenna (2007) implies that the damage is not confined to the moments being matched, since truncation also corrupts the mapping from reduced-form residuals to structural shocks that produced $\hat{\Psi}$ in the first place.

[**TBD:** Nota a pie' di pagina da sviluppare: in Christiano et al. (2005), Sezione IV, il polinomio del VAR stimato entra direttamente nel modello come processo dello shock di politica monetaria. ACEL stimano invece il blocco stocastico ζ_3 congiuntamente ai parametri strutturali. In entrambi i casi la struttura dei ritardi del VAR alimenta anche il lato modello, non solo il bersaglio: cambiando p si cambiano quindi due cose insieme. Verificare sul replication package come e' implementato in ACEL.]

6 Empirical Results

6.1 Data and specification

[**TBD:** Da scrivere: le dieci variabili del VAR di ACEL, il campione 1959:Q2–2001:Q4, le fonti e le trasformazioni. Precisare che uso il replication package degli autori, riferito alla versione NBER working paper del 2005 (Altig et al., 2005). Motivare i tre lag order: $p = 4$ e' il baseline di ACEL, $p = 12$ il triplo, nello spirito di De Graeve and Westermark (2025). Descrizione dettagliata delle variabili in Appendice A.]

6.2 VAR impulse responses: monetary policy shock

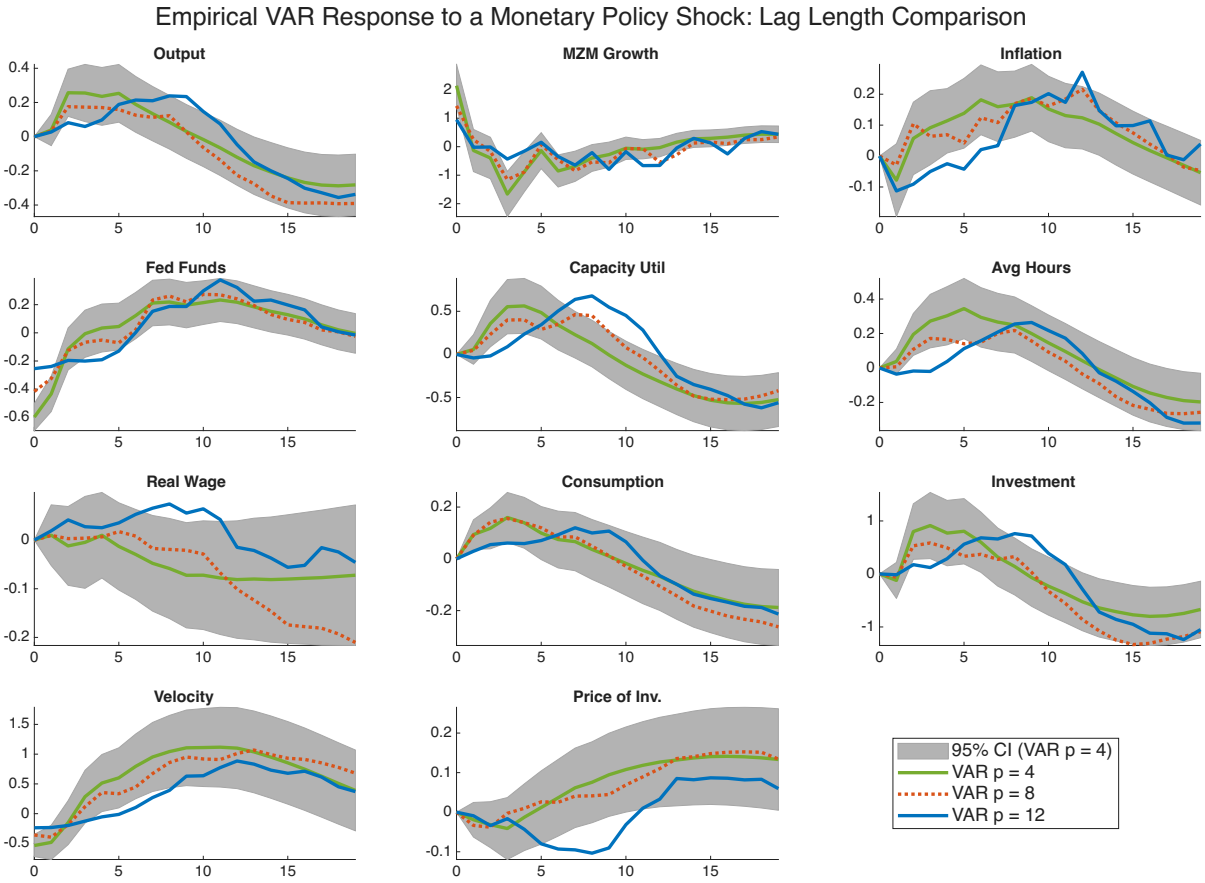


Figure 2: VAR impulse responses to a monetary policy shock: point estimates across lag orders against the VAR(4) confidence bands.

[**TBD:** Commentare la figura: segno e magnitudini restano stabili, ma le gobbe si formano e culminano con ritardo crescente al crescere di p, e le stime puntuali restano dentro l'intervallo di confidenza del VAR(4). Il punto sostanziale da sottolineare e' che la risposta dell'inflazione diventa piu' ritardata e piu' piatta a breve orizzonte: e' lo stesso fatto, visto dal lato delle IRF, del calo di $\hat{\gamma}$ documentato in Sezione 6.4. Cosi' la sezione IRF e la sezione parametri raccontano una catena unica invece di due osservazioni scollegate.

Riformulare inoltre il commento sul price puzzle, che allo stato attuale e' debole: la domanda binaria "c'e' o non c'e' il price puzzle" e' una distrazione rispetto al fatto rilevante, cioe' il ritardo crescente della risposta.]

6.3 Technology shocks: short-run versus long-run identification

[**TBD:** Sottosezione da promuovere dall'appendice al corpo. L'argomento: la politica monetaria e' identificata con restrizioni di breve periodo, gli shock tecnologici con restrizioni di lungo periodo. Queste ultime richiedono $\sum_i \Psi_i$, dove l'errore di troncatura si accumula,

e Ravenna (2007) predice quindi che le IRF identificate a lungo siano piu' fragili. Chari et al. (2008) sono la critica di riferimento proprio alle SVAR con restrizioni di lungo periodo. Il risultato empirico va in quella direzione: le IRF di politica monetaria restano dentro l'intervallo baseline, diverse di quelle tecnologiche no. E' un test di falsificazione interno che rende concreto il meccanismo invece di limitarsi ad asserirlo.

Inserire le due figure NT ed ET da Output/VAR_IRFs_Long_Lag_Progressively_Plots_4_8_12/. Prima pero' verificare se gli effetti di lungo periodo anomali su output, ore e investimenti a $p = 12$ siano un risultato o un problema di stima.]

6.4 DSGE structural parameters

Table 2: DSGE Structural Parameters Comparison Across Lag Lengths

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213	0.8213	0.8958
	(0.0479)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372	0.6448	0.8884
	(0.1503)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The Calvo price parameters $\xi_{p,hom}$ and $\xi_{p,fs}$ are not estimated directly: they are recovered numerically from the estimated structural parameters. Their standard errors are therefore obtained by Monte Carlo simulation with 1,000,000 draws from the joint distribution of the estimated structural parameters, solving for the implied Calvo parameter at each draw.

Three movements in the parameter estimates deserve comment.

The Phillips curve slope falls from 0.0402 at $p = 4$ to 0.0129 at $p = 12$. Equally important is what happens to its precision: at $p = 4$ and $p = 8$ the estimate is not

statistically distinguishable from zero, with standard errors of 0.0246 and 0.0598, while at $p = 12$ it is significant at the one percent level with a standard error of 0.0042. The long-lag VAR therefore delivers an estimate of γ roughly six times more precise, which is the variance-reduction mechanism of De Graeve and Westermark (2025) appearing inside the exercise itself. It also implies that ACEL's headline value is imprecisely estimated in their baseline specification.

The cost of varying capital utilisation collapses from 1.99 to 0.0042, three orders of magnitude. Given the discussion in Section 4, this is the movement driving the main result: with λ_f calibrated, σ_a is the only estimated parameter entering χ, and a vanishing σ_a flattens the firm's own marginal cost curve and pushes χ towards unity.

The investment adjustment cost roughly doubles, from 3.28 to 6.84, becoming precisely estimated only at $p = 12$. [**TBD:** Commentare: governa la gobba dell'investimento in CEE e ACEL, quindi e' una seconda dimensione lungo la quale le conclusioni strutturali cambiano con p. Verificare se il movimento e' coerente con il ritardo crescente delle gobbe nelle IRF.]

[**TBD:** Sull'inferenza per i parametri di Calvo: le distribuzioni Monte Carlo sono chiaramente non normali (bimodali a $p = 4$ e $p = 8$, con massa sopra $\xi_p = 1$ a $p = 8$) e i success rate sono 79%, 32% e 49%. Proposta: riportare intervalli percentili invece degli standard error, piu' il success rate, e spiegare che la non-normalita' discende dalla quadratica in (6), che diventa quasi singolare per $\gamma \to 0$. Dettagli in Appendice D. Da confermare con Efrem.]

6.5 Implied duration and the collapse of the firm-specific mechanism

Table 3: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR** ($p = 4$): Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

The Chi parameter

Table 4: Implied Strategic Complementarity Parameter χ Across Lag Lengths

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model (validity check, $\chi \equiv 1$)			
χ_{hom}	0.9989	1.0010	0.9977
	(0.0000)	(0.0000)	(0.0000)
Firm-Specific Capital Model			
χ_{fs}	0.0307	0.2030	0.8671
	(0.0596)	(0.1667)	(0.1698)

Note: χ is recovered by inverting $\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi$ at the estimated γ and implied ξ_p. $\chi_{hom} \equiv 1$ by construction (Proposition 1(i) in Altig et al., 2011); reported here only as a check that the numerical pipeline is consistent. Standard errors in parentheses are computed from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters' standard errors, applying the formula above draw-by-draw. The 90% Monte Carlo percentile interval for χ_{fs} is [0.0067, 0.0879], [0.0516, 0.5101], [0.3704, 0.9270] for $p = 4, 8, 12$ respectively; see the discussion of non-normality of the Calvo/chi Monte Carlo distributions in the text.

[**TBD:** Il risultato centrale del paper. A $p = 4$ si replica ACEL: $\xi_{p,fs} = 0.34$ contro $\xi_{p,hom} = 0.82$, cioe' 1.5 contro 5.6 trimestri, e solo il modello firm-specific riconcilia micro e macro. A $p = 12$ il divario sparisce: 0.89 contro 0.90, cioe' 9.0 contro 9.6 trimestri. Il modello firm-specific non riconcilia piu' nulla, perche' implica durate lunghe quanto quelle del modello omogeneo.

Spiegare poi *perche'* il divario collassa, che e' il punto teorico. Entrambi i parametri sono spinti verso uno dal calo di $\hat{\gamma}$, e la mappa $\gamma \to \xi_p$ e' compressiva in prossimita' di $\xi_p = 1$; ma la causa prima e' il crollo di σ_a, che azzera la complementarita' strategica e porta $\chi \to 1$. Chiudere con la citazione di ACEL p. 227: il VAR long-lag spinge la stima

esattamente nella regione in cui gli autori stessi ammettono che il meccanismo non regge.]

6.6 Residual diagnostics

[**TBD:** Tenere breve nel corpo, il resto in Appendice C. Riportare che RMSE dei residui e generalized variance scendono monotonicamente con p, ma dire esplicitamente che in-sample questo e' in buona parte meccanico e non costituisce prova di specificazione corretta. Sul test Portmanteau: non mi appoggio alla teoria asintotica, che con $K = 10$ e $p = 12$ non e' affidabile; ACEL stessi trovano p-value asintotici vicini a zero e p-value bootstrap tra l'87 e il 97 per cento.]

7 Conclusion

[**TBD:** Struttura proposta, tre paragrafi.

(1) Cosa faccio e cosa trovo. Replico ACEL arricchendo la struttura dei ritardi del VAR: le IRF identificate a breve periodo sono relativamente robuste, quelle identificate a lungo no; i parametri strutturali si spostano in modo sostanziale; il divario tra modello firm-specific e omogeneo scompare.

(2) Cosa significa. Non "ACEL avevano torto", ma che la loro conclusione dipende da una scelta — $p = 4$ — che non era percepita come una scelta sostanziale. Il meccanismo e' interno al loro stesso argomento, via σ_a e χ. Piu' in generale: IRFME e' presentata come robusta alla misspecificazione del modello, ma non e' robusta alla misspecificazione del VAR, e la seconda passa inosservata perche' tutti controllano la prima.

(3) Cosa resta aperto. Non sostengo che $p = 12$ sia il valore corretto: l'esercizio e' di sensibilita', non di selezione del modello. Estensioni: altri modelli stimati per IRF matching, e il confronto con le Local Projections alla luce di Ludwig (2026) e Plagborg-Møller and Wolf (2021).

Nota per me: le conclusioni sono la parte che sento piu' debole, da rivedere dopo il riscontro di Efrem.]

References

Altig, D., Christiano, L. J., Eichenbaum, M., and Lindé, J. (2005). Firm-specific capital, nominal rigidities and the business cycle. NBER Working Paper 11034, National Bureau of Economic Research. Working paper predecessor of the published version (Review of Economic Dynamics, 2011).

Altig, D., Christiano, L. J., Eichenbaum, M., and Lindé, J. (2011). Firm-specific capital, nominal rigidities and the business cycle. *Review of Economic Dynamics*, 14(2):225–247.

Antolín-Díaz, J., Fernández-Villaverde, J., and Guerrón-Quintana, P. (2026). Double descent in time series. Working paper, MIT, University of Pennsylvania, and Boston College.

Antolín-Díaz, J. and Surico, P. (2025). The long-run effects of government spending. *American Economic Review*, 115(7):2376–2413.

Aruoba, S. B. and Drechsel, T. (2024). Identifying monetary policy shocks: A natural language approach. Working paper, University of Maryland, NBER, and CEPR.

Barnichon, R., Mesters, G., Nakamura, E., and Steinsson, J. (2026). Humble vars. Working paper, University of California, Berkeley.

Baumeister, C. (2025). Discussion of "local projections or vars? a primer for macroeconomists". *NBER Working Paper Series*.

Canova, F. and Fosso, L. (2025). Low frequency movements and svar analyses. CAMP Working Paper 10/2025, Centre for Applied Macroeconomics and Commodity Prices (CAMP).

Chari, V. V., Kehoe, P. J., and McGrattan, E. R. (2008). Are structural vars with long-run restrictions useful in developing business cycle theory? *Journal of Monetary Economics*, 55(8):1337–1352.

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy*, 113(1):1–45.

Christiano, L. J., Eichenbaum, M., and Vigfusson, R. (2006). Assessing structural vars. *NBER Macroeconomics Annual*, 21:1–72.

De Graeve, F. and Westermark, A. (2025). Long-lag vars. *Journal of Monetary Economics*, 156.

Fernández-Villaverde, J., Rubio-Ramírez, J., and Sargent, T. J. (2005). A, b, c's (and d)'s for understanding vars. NBER Technical Working Paper 308, National Bureau of Economic Research.

Forni, M., Gambetti, L., and Sala, L. (2016). Var information and the empirical validation of dsge models. RECent Working Paper 119, RECent - Center for Economic Research, University of Modena and Reggio Emilia.

Giacomini, R. (2013). The relationship between dsge and var models. cemmap Working Paper CWP21/13, Centre for Microdata Methods and Practice (cemmap).

Jordà, O., Singh, S. R., and Taylor, A. M. (2024). The long-run effects of monetary policy. *The Review of Economics and Statistics*.

Li, D., Plagborg-Møller, M., and Wolf, C. K. (2024). Local projections vs. vars: Lessons from thousands of dgps. Working paper, Two Sigma Investments, Princeton University, and MIT.

Ludwig, J. F. (2026). Local projections are var predictions of increasing order. *Working Paper, R&R Econometrica 2nd Round*.

Olea, J. L. M., Plagborg-Møller, M., Qian, E., and Wolf, C. K. (2025). Local projections or vars? a primer for macroeconomists. NBER Working Paper 33871, National Bureau of Economic Research.

Plagborg-Møller, M. and Wolf, C. K. (2021). Local projections and vars estimate the same impulse responses. *Econometrica*, 89(2):955–980.

Ramey, V. A. (2016). Macroeconomic shocks and their propagation. In *Handbook of Macroeconomics*, volume 2, pages 71–162. Elsevier.

Ravenna, F. (2007). Vector autoregressions and reduced form representations of dsge models. *Journal of Monetary Economics*, 54(7):2048–2064.

Rossi, B., Hall, A. R., Inoue, A., and Nason, J. M. (2012). Information criteria for impulse response function matching estimation of dsge models. *Journal of Econometrics*, 170(2):499–518.

Smets, F. and Wouters, R. (2007). Shocks and frictions in US business cycles: A Bayesian DSGE approach. *American Economic Review*, 97(3):586–606.

Theodoridis, K. (2011). An efficient minimum distance estimator for dsge models. Bank of England Working Paper 439, Bank of England.

A Data

[**TBD:** Tabella con le dieci variabili: definizione, fonte, trasformazione, periodo campionario. Sul modello della Table A.1 di Lami.]

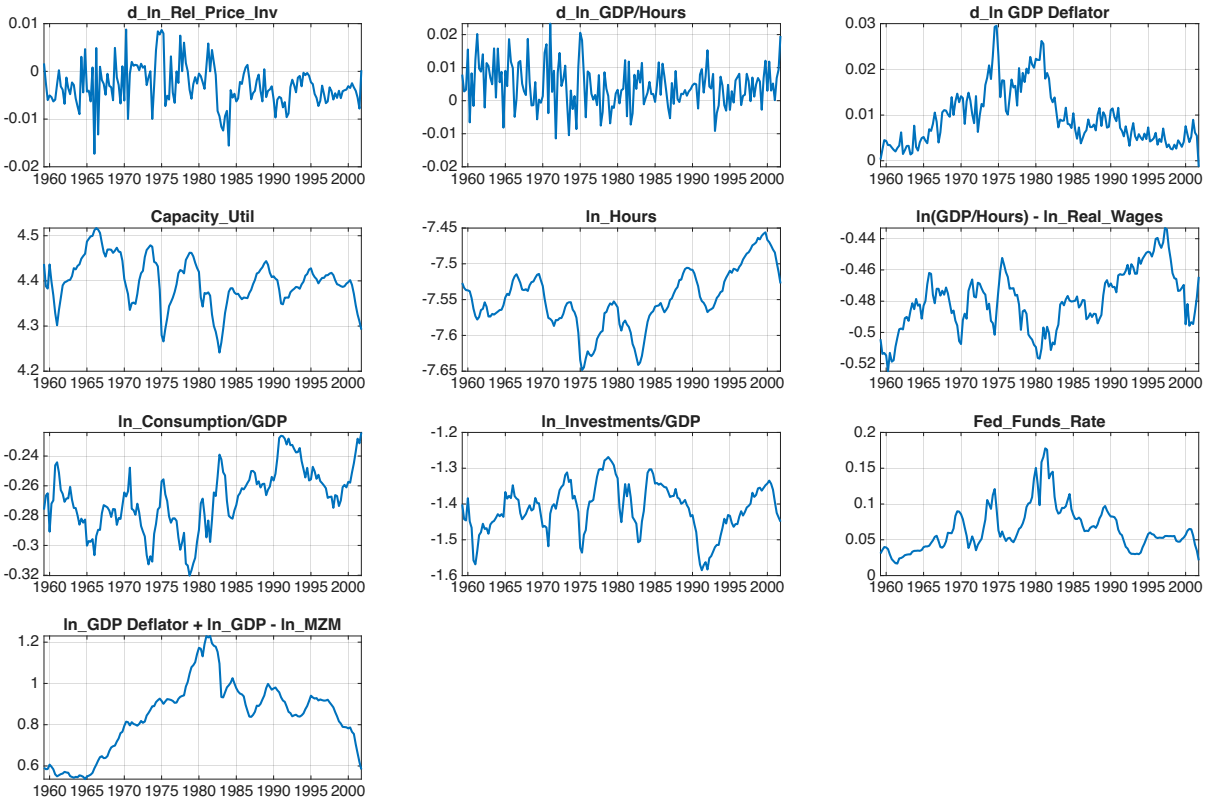


Figure 3: Raw historical series

B The ACEL Model: Full Derivation

[**TBD:** Derivazione completa: famiglie, imprese, aggregazione alla Yun (1996), politica monetaria, condizioni di equilibrio log-linearizzate. Piu' la derivazione esplicita di χ come funzione dei parametri strutturali.]

C Additional Residual Analysis

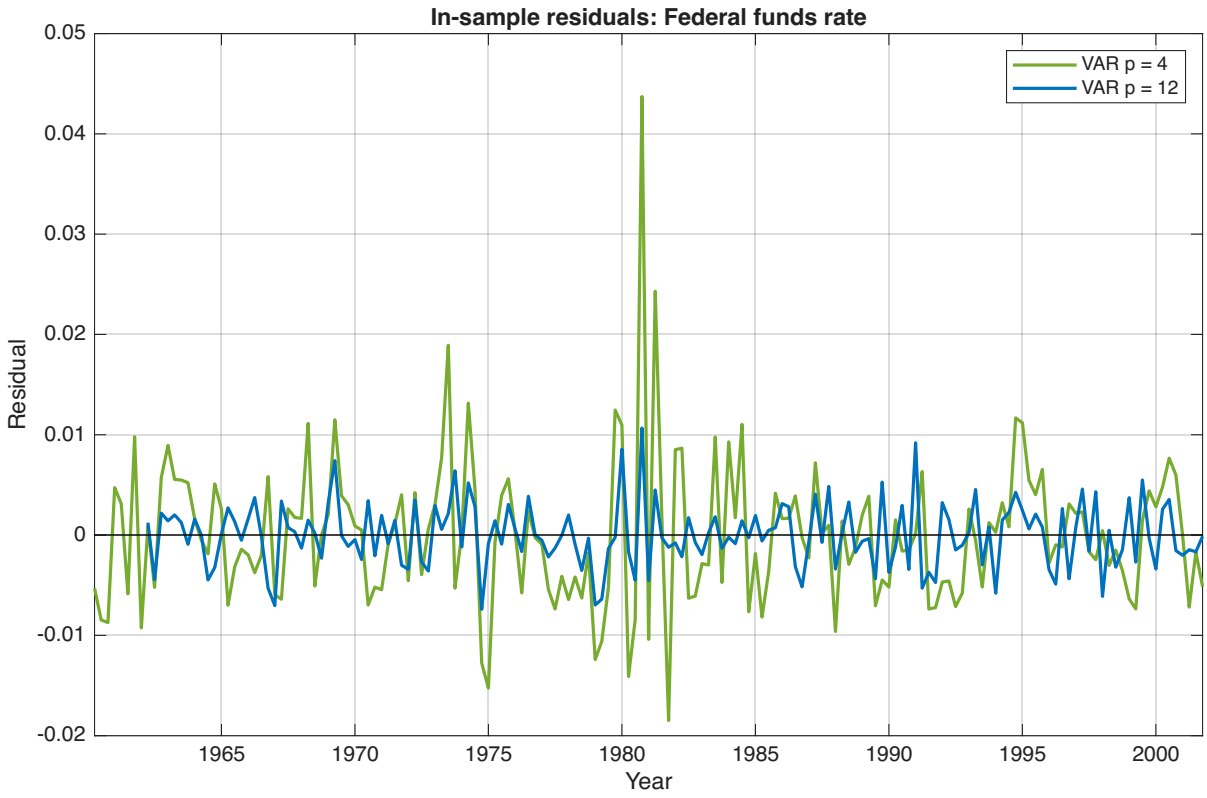


Figure 4: In-sample residuals: federal funds rate, $p = 4$ versus $p = 12$.

Table 5: In-sample RMSE of the reduced form VAR residuals

Variable	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Δ ln rel. price of investment	0.3387	0.2752	0.1962
Δ ln GDP / hours	0.5220	0.3758	0.2470
Δ ln GDP deflator	0.2095	0.1823	0.1190
Capacity utilisation	1.1726	0.8427	0.5110
ln hours	0.4977	0.4032	0.2410
ln(GDP/hours) - ln real wage	0.5469	0.4629	0.3278
ln consumption / GDP	0.5690	0.4374	0.3059
ln investment / GDP	1.8667	1.4188	0.9546
Federal funds rate	0.7179	0.5340	0.3201
Velocity: ln P + ln GDP - ln MZM	1.4146	1.0896	0.7643

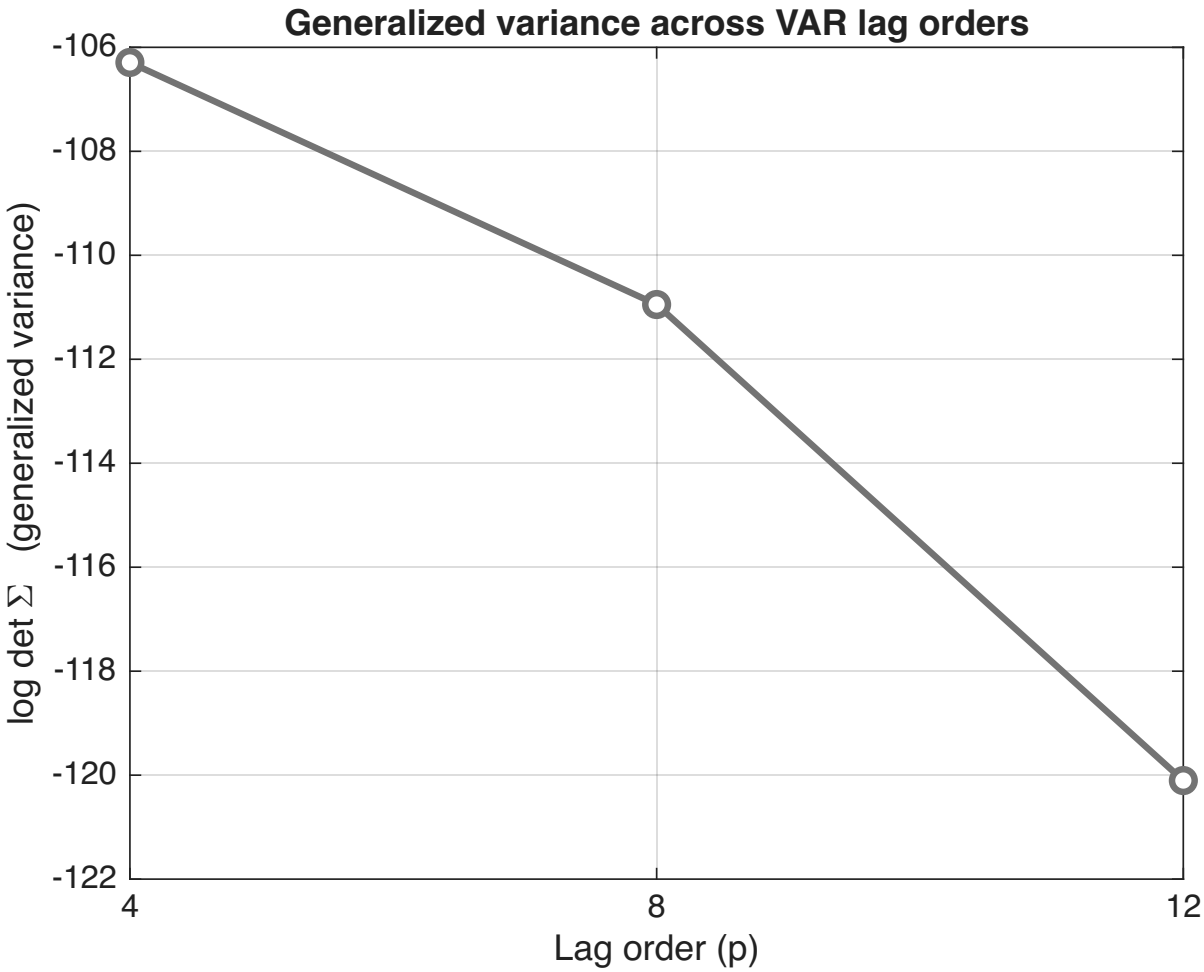


Figure 5: Generalized variance across lag orders

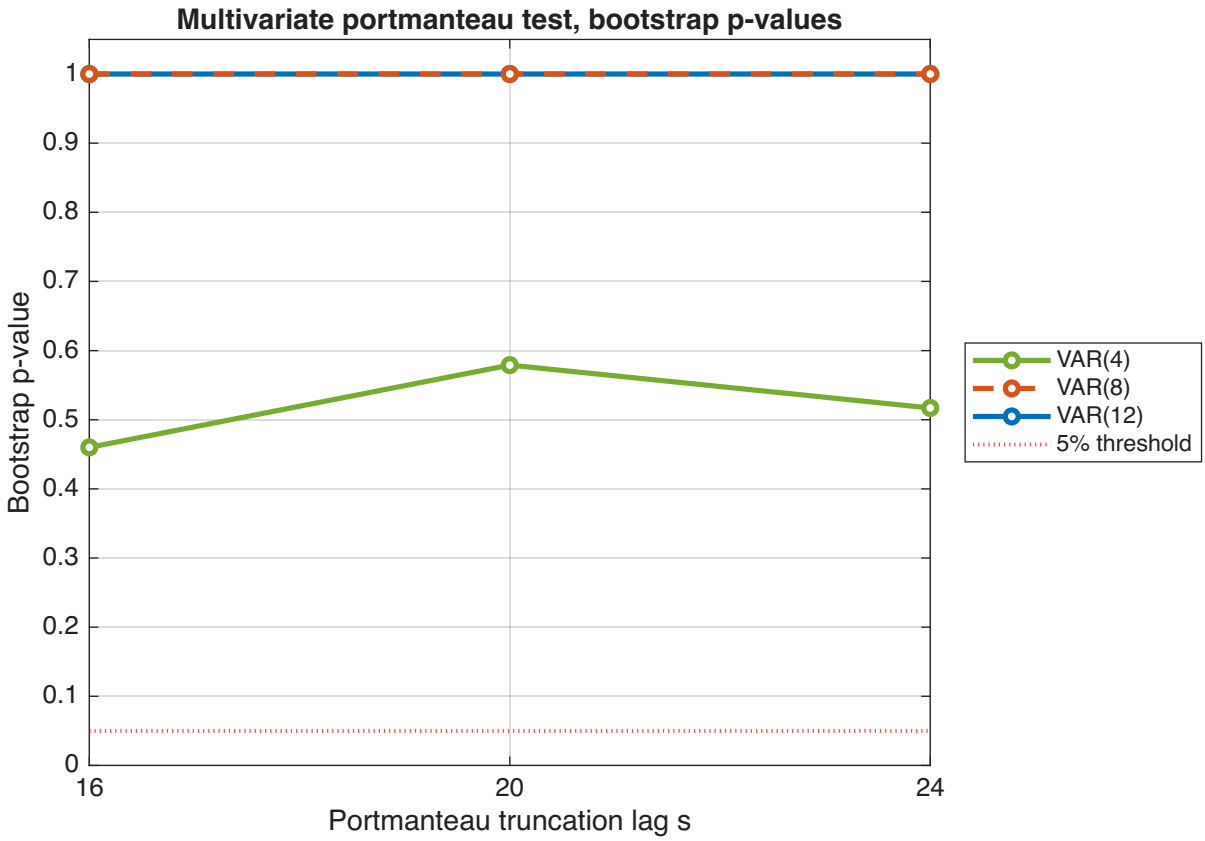


Figure 6: Portmanteau p-values across lag orders



Figure 7: Standard deviation ratios (reduced form)

D Monte Carlo Inference on the Calvo Parameters

[**TBD:** Descrivere la simulazione: un milione di estrazioni dalla distribuzione congiunta dei parametri strutturali stimati, mappate attraverso (6) per ottenere ξ_p nei due modelli. Riportare le tre densita' e i success rate (79%, 32%, 49%). Spiegare perche' il metodo delta e l'assunzione di normalita' congiunta usata nella stima non sopravvivono alla trasformazione.]

E Robustness

[**TBD:** Candidati: campione comune per tutti i lag order; lag order intermedi; specificazioni alternative dell'identificazione; sensibilita' agli orizzonti matchati.]

F Replication Files and Code

[**TBD:** Link al repository GitHub con i codici MATLAB e i file di output.]